KARAN KAHER

Data Engineer - AI & Data

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Professional Summary

Data Engineer with 5+ years of experience delivering end-to-end data pipelines, cloud warehouses, and BI reporting systems. Proven track record of diagnosing and resolving complex SQL performance bottlenecks for enterprise clients. Uniquely combines traditional data engineering with applied Generative AI system design to build intelligent, production-grade workflows and AI personas.

Technical Skills

Languages	Python, SQL, PySpark
Data Engineering	Apache Airflow, Apache Kafka, dbt, SSIS, Databricks
Data Modeling / Warehousing	Snowflake, AWS Redshift, SQL Server, DB2, Data Modeling, ETL/ELT Design
Cloud Platforms	Microsoft Azure (ADF, Synapse Analytics), AWS (Redshift, S3, EC2)
DevOps & Infrastructure	Docker, Kubernetes, Terraform, Jenkins, GitLab CI/CD, GitHub Actions
Generative AI	RAG, LLM Agents, Prompt Engineering, Vector Search, Claude Code
BI & Visualization	Power BI, Tableau, SSRS, Lookerstudio

Professional Experience

Software Engineer – Data & Analytics **Feb 2021 – Present**

Damco Solutions Private Limited (IT Services & Consulting) *Remote, India*

- Engineered and orchestrated modern data pipelines using **Apache Airflow** and **PySpark** on **Databricks**, extracting high-volume data from on-premise systems and landing it into **Snowflake** to support scalable cloud analytics.
- Redesigned complex **SSIS ETL packages** and built hybrid pipelines using **Azure Data Factory (ADF)** to extract policy and finance data from legacy systems (**SQL Server, DB2**). Replaced inefficient full-table reloads with robust **incremental/delta extraction logic**, **meaningfully reducing nightly batch processing runtime**.
- Diagnosed severe performance bottlenecks in legacy SSRS reporting systems by analyzing **SQL Server execution plans**; rewrote 40+ complex queries—utilizing **CTEs, window functions, and covering indexes**—**cutting average execution time by up to 60%** across 92 enterprise reports.
- Restructured the foundational claims and premium **data models** for **AML compliance**, resolving a recurring data-mismatch bug by correcting inconsistent join grains between **fact and dimension tables** and ensuring accuracy in suspicious-transaction reporting.
- Partnered directly with global actuarial and finance stakeholders to translate ambiguous requirements into self-service **Power BI dashboards** and **Lookerstudio**, replacing single-threaded SQL batch processes and **reducing recurring ad-hoc reporting requests**.
- Established team-wide **engineering standards** by authoring comprehensive **SOPs** covering SQL query optimization patterns, dataset caching, and ETL parameter design, **shortening new-developer ramp-up time**.

Key Projects

Independently delivered as the assigned engineer on each client engagement below.

Retail/FMCG – Cloud Data Integration & Analytics Platform (Snowflake, Airflow, dbt)

- Designed **Airflow DAGs** to orchestrate multi-source extraction, built **dbt models** to standardize and transform the data, and modeled a **star schema** in **Snowflake** for downstream analytics.
- Replaced manual spreadsheet reconciliation with a single automated pipeline, improving data freshness from a weekly manual refresh to **daily automated loads** and giving analysts a trusted, query-ready dataset.

Finance – Enterprise Data Lakehouse & Reconciliation (Azure, Databricks, PySpark)

- Architected a Medallion architecture (Bronze/Silver/Gold) utilizing **Azure Data Factory (ADF)** for ingestion into **ADLS** and authored complex **PySpark** jobs on **Databricks** for deduplication and SCD Type 2 logic.
- Modernized the analytics infrastructure, substantially cutting daily reconciliation processing time and providing a single, highly performant source of truth for downstream Power BI reporting.

Insurance – Streaming Pipelines & Cloud Data Warehouse (AWS, Kafka, Redshift)

- Built a **Kafka** event pipeline on **EC2** to stream claims in near real time into **AWS S3** and orchestrated ETL pipelines to load normalized dimension tables into **AWS Redshift**, tuning physical design keys to match query patterns.
- Cut the fraud-detection lag from a 24-hour batch cycle to near real time, while noticeably improving average analytical query runtimes and reducing legacy on-premise technical debt.

Data Infrastructure & DevOps Automation (Terraform, Kubernetes, CI/CD)

- Containerized ETL applications with **Docker**, defined cloud infrastructure as code via **Terraform**, and built CI/CD pipelines using **GitLab CI** / **GitHub Actions** to automate deployment onto **Kubernetes** clusters.
- Eliminated manual provisioning bottlenecks, ensuring reproducible environments and reliable, **zero-downtime releases**.

Generative AI – AI Decision Due Diligence Analyst

- Designed an AI "Due Diligence Analyst" with a defined **RAG workflow** (semantic chunking, **vector similarity search**) to classify queries, retrieve policy evidence, cross-check against rules, and produce recommendations.
- Constrained responses to a **fixed JSON schema** for downstream tools and enforced behavioral rules to cite sources and avoid hallucinations, drastically reducing manual document lookup time for business teams.

Education

B.Tech, Computer Science & Engineering **2016 – 2020**

Manav Rachna University *CGPA: 8.88*